

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
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Section / Batch:	CSE-AI/2024-2028	Date:	21-08-2026

Code of Conduct

I Malli Chandana(192472250)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student : Malli Chandana

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

1. Recall Comparison in Detection

Results:

MODEL	TP	FN
A	60	40
B	80	20

(a) Compute recall.

(b) Compare and evaluate sensitivity.

Introduction

In object detection, image classification, and medical diagnosis systems, it is important to measure how effectively a model identifies the actual positive cases. One of the most important evaluation metrics for this purpose is Recall, also called Sensitivity or True Positive Rate.

“Out of all the actual positive cases, how many did the model correctly detect?”

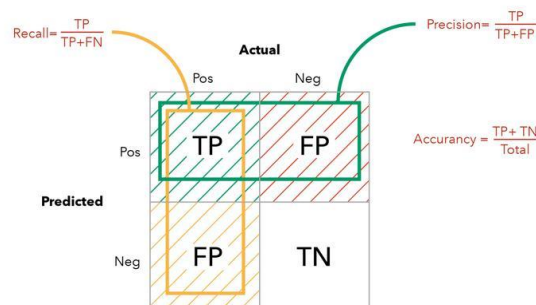
A model with high recall misses fewer positive cases. Therefore, recall is particularly important in applications such as disease detection, security systems, face detection, and safety-critical object detection, where missing a positive case can have serious consequences.

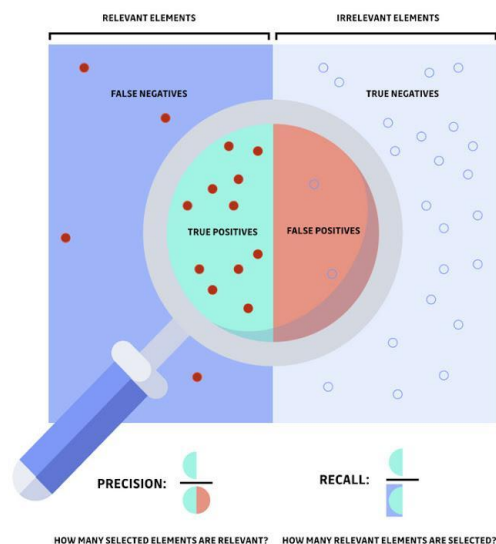
For the given problem, two detection models are compared:

Model	True Positive (TP)	False Negative (FN)
A	60	40
B	80	20

Here:

- **True Positive (TP):** Positive cases correctly detected by the model.
- **False Negative (FN):** Actual positive cases that the model failed to detect.





(a) Computation of Recall

The formula for recall is:

Recall is generally expressed as a percentage.

Recall of Model A

For Model A:

- $TP = 60$
- $FN = 40$

Therefore,

Converting into percentage:

This means that **Model A correctly detects 60% of all actual positive cases**. The remaining 40% are missed by the model.

Recall of Model B

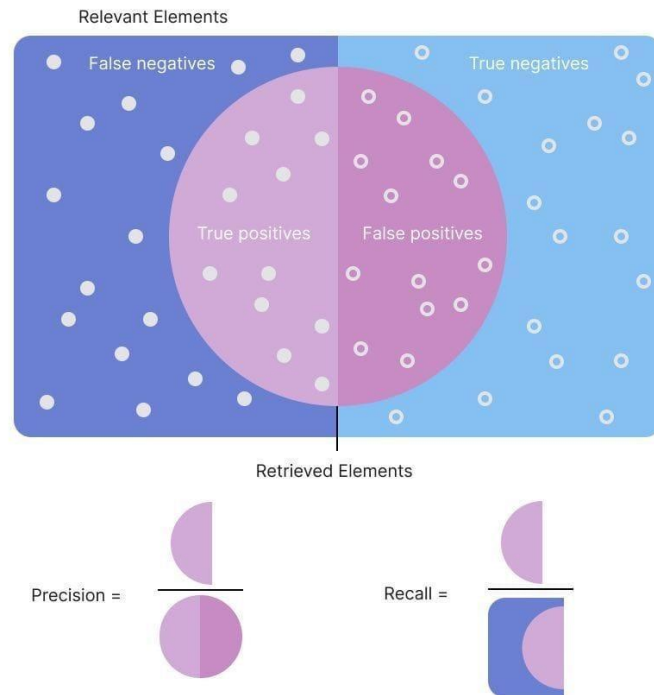
For Model B:

- $TP = 80$
- $FN = 20$

Therefore,

Converting into percentage:

This means that **Model B correctly detects 80% of all actual positive cases**, while only 20% are missed.



$$\text{Precision} = \frac{TP}{TP + FP}$$

"Of all predicted churn users, how many were correct?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

"Of all actual churn users, how many were correctly identified?"

Where:

- TP (True Positives): Correctly predicted churners
- FP (False Positives): Incorrectly predicted as churners
- FN (False Negatives): Missed actual churners

(b) Comparison and Evaluation of Sensitivity

Recall and sensitivity refer to the same performance measure. They indicate how well a model identifies actual positive cases.

The calculated values are:

Performance Measure	Model A	Model B
True Positives	60	80
False Negatives	40	20
Recall	60%	80%
Sensitivity	60%	80%

From the table, it is clear that Model B performs better than Model A in terms of sensitivity.

Model B achieves a recall of 80%, whereas Model A achieves only 60%.

The difference is:

Therefore, Model B has a 20 percentage-point higher recall than Model A.

Another important observation is the number of false negatives. Model A has 40 false negatives, while Model B has only 20 false negatives.

Thus, Model B misses 20 fewer positive cases.

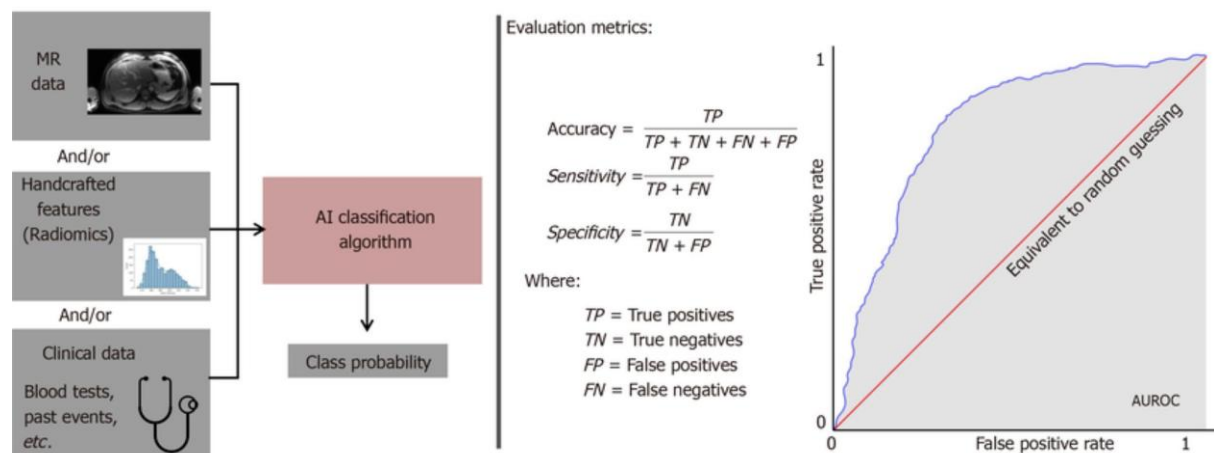
Why is this important?

Consider a medical detection system. Suppose the positive case represents a patient who actually has a particular disease.

- If the system correctly identifies the disease → True Positive
- If the system fails to identify the disease → False Negative

A high number of false negatives can be dangerous because actual patients may be incorrectly considered healthy.

Similarly, in an object detection system, a false negative occurs when an object that is actually present is not detected. For example, an autonomous vehicle failing to detect a pedestrian could create a serious safety problem.



Interpretation of the Results

Model A

Model A has:

This indicates that Model A successfully detects **6 out of every 10 actual positive cases**.

However, it misses approximately **4 out of every 10 positive cases**. Therefore, its sensitivity is comparatively lower.

Model B

Model B has:

This indicates that Model B successfully detects **8 out of every 10 actual positive cases**.

It misses only **2 out of every 10 positive cases**, making it more effective when the goal is to minimize missed detections.

Practical Comparison

The choice between the models depends on the application.

If the application requires **high sensitivity**, Model B is clearly preferable.

For example:

- **Medical diagnosis:** Missing a disease case can be dangerous.
- **Security surveillance:** Missing a suspicious object or person may create a security risk.
- **Autonomous vehicles:** Missing a pedestrian or obstacle can cause accidents.
- **Fire detection:** Missing an actual fire can have severe consequences.
- **Industrial inspection:** Missing a defective product can affect quality and safety.

In all these situations, a model with fewer false negatives is generally preferred.



Conclusion

Recall, also known as **Sensitivity**, measures the ability of a detection model to correctly identify actual positive cases.

For **Model A**:

For **Model B**:

Therefore:

in terms of recall and sensitivity.

Model B is the better model because it correctly detects a greater proportion of actual positive cases and produces fewer false negatives. Its sensitivity is 20 percentage points higher than Model A. Hence, when the main objective is to minimize missed detections, Model B should be selected.

2. Depth Estimation Error (Squared)

Predicted:

Object A B Actual

1 2.5 2.2 2

2 3.5 3.1 3

3 4.5 4.2 4

(a) Compute squared error.

(b) Compare and evaluate accuracy.

Introduction

Depth estimation is an important task in computer vision where a system estimates the distance or depth of an object from a camera. It is widely used in autonomous vehicles, robotics, 3D reconstruction, augmented reality, and object detection.

In this problem, two models, Model A and Model B, predict the depth of three objects. These predictions are compared with the actual depth values.

Object	Model A Prediction	Model B Prediction	Actual Depth
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

The objective is to calculate the **squared error** for each prediction and determine which model provides better depth estimation accuracy.

(a) Computation of Squared Error

The **squared error** is calculated using:

A smaller squared error indicates that the predicted value is closer to the actual value.

Object 1

Actual depth = **2**

Model A

Predicted depth = 2.5

Model B

Predicted depth = 2.2

Therefore, for Object 1, **Model B has a smaller error.**

Object 2

Actual depth = **3**

Model A

Predicted depth = 3.5

Model B

Predicted depth = 3.1

Therefore, for Object 2, **Model B again has a smaller error.**

Object 3

Actual depth = **4**

Model A

Predicted depth = 4.5

Model B

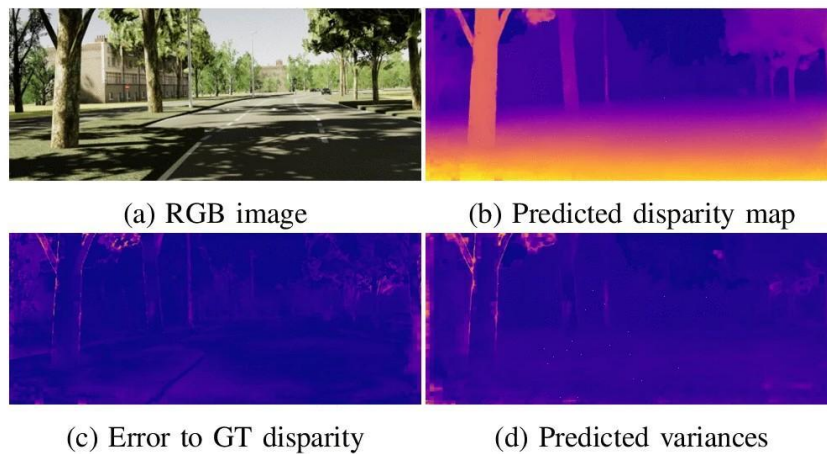
Predicted depth = 4.2

Again, **Model B has a smaller error.**

Squared Error Comparison

The complete calculation can be summarized as follows:

Object	Actual	Model A	A Squared Error	Model B	B Squared Error
1	2.0	2.5	0.25	2.2	0.04
2	3.0	3.5	0.25	3.1	0.01
3	4.0	4.5	0.25	4.2	0.04
Total			0.75		0.09



Mean Squared Error

To compare the overall performance of the two models, we can calculate the **Mean Squared Error (MSE)**.

The formula is:

Model A

Total squared error:

Therefore:

Model B

Total squared error:

Therefore:

(b) Compare and Evaluate Accuracy

The MSE results are:

Model	Total Squared Error	Mean Squared Error
Model A	0.75	0.25
Model B	0.09	0.03

Since **lower MSE means better prediction accuracy**, Model B performs significantly better than Model A.

Model A

Model A predicts:

- 2.5 instead of 2
- 3.5 instead of 3
- 4.5 instead of 4

The prediction error is consistently **0.5** for all three objects. Consequently, the squared error is **0.25** for every object.

Model A therefore has:

Model B

Model B predicts:

- 2.2 instead of 2
- 3.1 instead of 3
- 4.2 instead of 4

These predictions are much closer to the actual depth values. Its errors are only **0.2, 0.1, and 0.2**, respectively.

Therefore:

Model B has a much smaller MSE.

Practical Significance

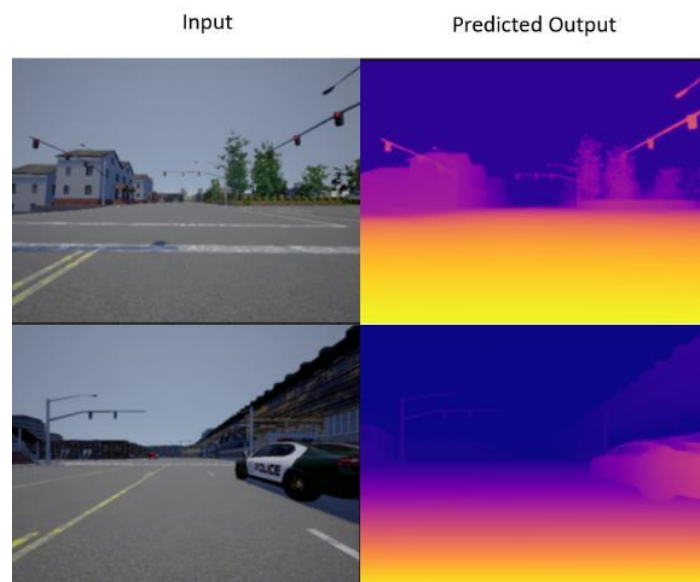
Accurate depth estimation is particularly important in **autonomous driving**. An autonomous vehicle needs to estimate how far objects such as pedestrians, vehicles, and obstacles are from the vehicle.

For example, if the actual distance of an obstacle is **3 metres**:

- Model A predicts **3.5 m**
- Model B predicts **3.1 m**

Model B provides an estimate that is much closer to reality. More accurate depth estimation can help an autonomous system make better decisions regarding **braking, steering, obstacle avoidance, and safe navigation**.

Depth estimation is also useful in robotics. A robot can use depth information to determine how far an object is from its robotic arm and calculate the appropriate movement required to pick it up.



Conclusion

The squared errors for the three objects are:

Model A

Total squared error:

Mean Squared Error:

Model B

Total squared error:

Mean Squared Error:

Therefore, **Model B is more accurate than Model A** because it has a much lower squared error and MSE. Its predictions are consistently closer to the actual depth values.

15. Recognition Precision Comparison

Results:

System TP FP

A 90 10

B 85 5

(a) Compute precision.

(b) Compare and evaluate performance.

Introduction

Precision is an important evaluation metric used in image recognition, object detection, face recognition, medical diagnosis, and other machine learning systems. It measures how many of the cases identified as **positive by the system are actually positive**.

In simple words, precision answers:

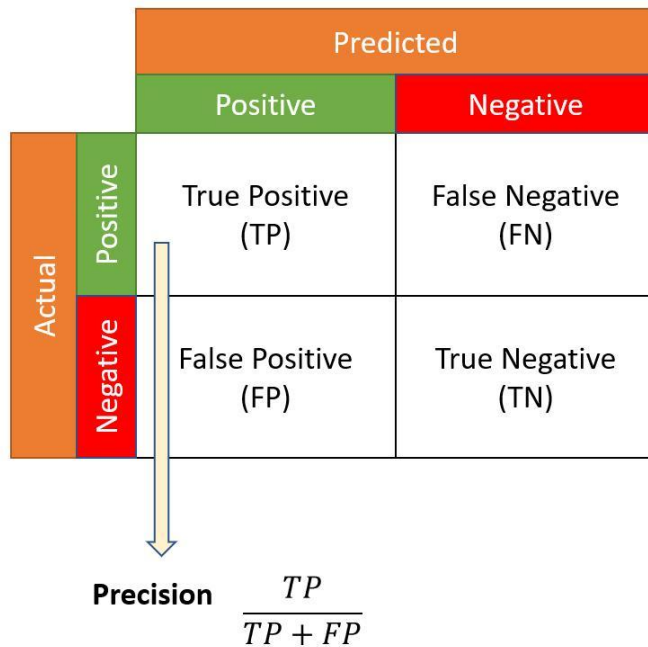
“Out of all the objects the system recognized as positive, how many were actually correct?”

The given results are:

System	True Positive (TP)	False Positive (FP)
A	90	10
B	85	5

Here:

- **True Positive (TP):** Correctly recognized positive cases.
- **False Positive (FP):** Cases incorrectly recognized as positive.



(a) Compute Precision

The formula for **Precision** is:

A higher precision means that the system produces **fewer incorrect positive predictions**.

Precision of System A

For System A:

- TP = 90
- FP = 10

Substituting the values:

Converting into percentage:

This means that out of every 100 cases recognized as positive by System A, approximately **90 cases are actually positive**, while 10 are incorrect positive predictions.

Precision of System B

For System B:

- TP = 85
- FP = 5

Therefore:

Converting into percentage:

This means that approximately **94.44% of the positive predictions made by System B are correct.**

(b) Compare and Evaluate Performance

The calculated precision values are:

Performance Measure	System A	System B
True Positive (TP)	90	85
False Positive (FP)	10	5
Precision	90%	94.44%

From the results, **System B has higher precision** than System A.

The difference in precision is:

Therefore, System B has approximately **4.44 percentage points higher precision.**

Analysis of System A

System A correctly recognizes **90 positive cases**, which is higher than System B's 85 true positives.

However, System A also produces **10 false positives.**

This means that although System A recognizes more positive cases correctly, it also makes more incorrect positive predictions.

Its precision is:

Therefore, System A provides good recognition performance but has a relatively higher rate of false positive predictions.

Analysis of System B

System B correctly recognizes **85 positive cases** and produces only **5 false positives.**

Although its number of true positives is slightly lower than System A, it makes **half as many false positive errors.**

Its precision is:

Therefore, System B is more reliable when the goal is to ensure that **positive predictions are correct.**

Why False Positives Matter

False positives can be problematic in many real-world recognition applications.

For example, consider a **security recognition system**:

- Correctly identifying a suspicious person → **True Positive**
- Incorrectly identifying an innocent person as suspicious → **False Positive**

If a system produces many false positives, security personnel may need to investigate many unnecessary alerts.

Similarly, in **object detection**, a false positive could mean that the system identifies an object where no such object exists.



Therefore, higher precision is valuable when **incorrect positive predictions are costly or undesirable**.

Overall Comparison

System A has a higher number of correct positive predictions:

while System B has:

However, System A also has more false positives:

compared with:

This results in:

and

Thus, **System B provides better precision**, even though it has fewer true positives.

Conclusion

The precision of the two recognition systems is:

Therefore:

System B is preferable when the main objective is to **reduce false positive predictions and increase the reliability of positive recognition results**. Although System A detects 5 more true positives, System B produces 5 fewer false positives and achieves approximately **94.44% precision**, making it the more precise and reliable system for positive predictions.